Curated Research Articles

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- **Optical nanoscopy of spatiotemporal metal stripping cooperativity at single-ion and sub-particle resolution** — score: 1.000 The article presents a novel ion-localization optical nanoscopy technique that achieves single-ion imaging with high spatial (50 nm) and temporal (20 ms) resolution. This advancement allows researchers to investigate the complex interfacial processes during battery operation, specifically observing the cooperative behavior of metal stripping on zinc anodes, which is essential for understanding battery dynamics. Journal: *Nature Materials*
- **Beyond Bulk Transport: Interfacial Atomic Regulation for Practical Ah-Scale Zinc Anodes** — score: 1.000 The study presents a novel strategy to enhance zinc deposition by utilizing a viscous additive that minimizes bulk ion diffusion while promoting rapid surface transport, thereby preventing dendrite formation. This approach enables the development of stable and high-capacity zinc electrodes, achieving over 7800 hours of stability and a record discharge capacity in battery cells, paving the way for more efficient and durable aqueous zinc batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Redox Complexity in Sodium Manganese Hexacyanomanganate and Its Influence on Sodium-Ion Storage Performance** — score: 1.000 This study investigates the redox mechanisms in sodium manganese hexacyanomanganate (NaMnHCMn), finding that only low-spin manganese at the carbon site undergoes redox changes, while high-spin manganese remains inactive. This exclusive involvement of MnC sites in all three redox processes suggests a unique behavior compared to other sodium Prussian blue analogues and reveals an identified Mn1+ intermediate with a distinct infrared signature, contributing to the material's limited cycling stability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Ultra-Low-Strain Calcium and Magnesium Ion Storage Enabled by Tunnel-Structured MoO3 Positive Electrode** — score: 1.000 The article presents a novel hexagonal tunnel-structured MoO3 material engineered for use as a positive electrode in calcium and magnesium batteries. This innovative nano-h-MoO3 demonstrates exceptional capacity and structural stability during divalent cation intercalation, with minimal lattice distortion, offering a promising solution for enhancing the energy density and efficiency of next-generation divalent metal batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Atomic-Scale Imaging of Li+ Trapping at Defects in Degraded LiCoO2** — score: 1.000 The article presents atomic-scale imaging findings on lithium ion (Li+) trapping at defects in degraded lithium cobalt oxide (LiCoO2) materials. This research highlights how such defects influence the electrochemical performance of LiCoO2, which is significant for improving lithium-ion battery technology. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Advanced solid-state NMR for cathode materials: Insights into local structure and ionic dynamics mechanisms in lithium- and sodium-ion batteries** — score: 1.000 The article discusses advancements in solid-state Nuclear Magnetic Resonance (NMR) techniques to investigate the local structural properties and ionic dynamic mechanisms of cathode materials used in lithium- and sodium-ion batteries. These insights potentially enhance the understanding and performance of energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Unveiling the High-Voltage Reactivity and Gas Evolution With Aluminum-Based Chloride and Oxychloride Catholytes in Solid-State Sodium Batteries** — score: 1.000 This study investigates the electrochemical performance of crystalline NaAlCl4 and amorphous sodium–aluminum–oxychloride solid electrolytes in all-solid-state sodium batteries using NaNi0.5Mn0.5O2 cathodes. The findings reveal that, while the oxychloride improves ionic conductivity, it suffers from interfacial instability and gas evolution at high voltages, making pure chloride electrolytes like NaAlCl4 a safer and more stable option for high-voltage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Operando monitoring reveals the critical role of cut-off voltage in the mechanochemical degradation of NCM811** — score: 1.000 This study investigates how cut-off voltage influences the mechanochemical degradation of NCM811 battery materials through operando monitoring techniques. The findings underscore the importance of voltage parameters in optimizing material stability and performance during energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Asymmetric Multi-Site Ion Exchange in Porous Carbon Electrodes** — score: 1.000 The study introduces a novel 2D EXSY NMR framework to analyze ion dynamics in porous carbon electrodes, addressing the complexities of ion transport through heterogeneous pore structures. By employing a lognormal rate distribution, it captures varying exchange kinetics, revealing insights into ion mobility and diffusion characteristics across different pore architectures. These findings enhance understanding of ion transport in energy storage and separation technologies, paving the way for optimized porous carbon design. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Building a Long-Cycle-Life Rechargeable Daniel Cell from Solvent Isotope Effect of Hydrogen** — score: 0.900 The study introduces a rechargeable Zn–Cu battery utilizing heavy-water electrolytes, which enhance the electroplating process via the hydrogen solvent isotope effect. This effect reduces desolvation energy and promotes uniform metal deposition, resulting in superior battery performance with extended lifespan and high efficiency, marking a significant advancement in sustainable energy storage technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Taming Free Water Activity via Ethylene Glycol for Durable and Fast-Charging Rechargeable Seawater Batteries** — score: 0.800 The article presents a novel approach to enhancing the performance of aqueous seawater batteries by using ethylene glycol to modify seawater, which decreases free water activity and improves Na+ mobility. This modification leads to exceptional battery stability, achieving over 20,000 cycles with 96.50% capacity retention and rapid charge/discharge capabilities, positioning this technology as a promising solution for sustainable marine energy storage. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergistic Current Collector and Electrolyte Additive Enable 10 000-Cycle Zinc Iodine Batteries with Low N/P Ratios** — score: 0.800 The article presents a novel approach to improve the lifespan of aqueous zinc-iodine batteries with low negative-to-positive (N/P) ratios by using a graphene-coated copper current collector and a polyquaternary ammonium salt (PUB) electrolyte additive. This design significantly reduces hydrogen evolution and mitigates the polyiodide shuttle effect, allowing the battery to achieve an impressive 10,000 cycles with a 172-fold increase in longevity compared to traditional cells. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Upcycling of Spent High-Nickel Layered Cathodes: A Mechanism-Guided Framework for Single-Crystal Structural Healing** — score: 0.800 This review presents a novel framework for the upcycling of spent high-nickel layered cathodes, emphasizing the transformation from simple recovery to advanced structural regeneration through single-crystal architecture. It details the degradation mechanisms of cathodes and advocates for the integration of AI diagnostics and sustainability metrics to promote a high-value, closed-loop battery economy, addressing challenges in conventional recycling methods. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Cathode-Confined Polysulfide Retention-Release Reprograms Li2S Deposition in High-Loading Li–S Batteries** — score: 0.800 The study investigates a novel cathode design that enhances the retention and release of polysulfides in high-loading lithium-sulfur (Li-S) batteries. This reprogramming significantly influences the deposition of lithium sulfide (Li2S), leading to improved battery efficiency and longevity. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **Functionalized Hydrated Eutectic-Like Electrolyte Boosting Interfacial Compatibility for Ultra-Stable Aqueous Potassium-Ion Batteries** — score: 0.800 The study introduces a high-performance aqueous potassium-ion battery (AKIB) utilizing a specially designed hydrated eutectic-like electrolyte, which enhances ion solvation and expands the electrochemical stability window. Coupled with a dual-metal Prussian blue cathode, the battery exhibits remarkable electrochemical performance,

including stable voltage output, high capacity retention, and excellent cycling stability. This electrolyte-driven strategy aims to optimize the compatibility and efficiency of AKIBs for large-scale energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **[ASAP] Energy-Transfer-Modulated Structural Evolution during Lithium–Sodium Ion Exchange in Layered Oxide Cathodes** — score: 0.800 The article investigates how energy transfer influences the structural changes in layered oxide cathodes during the exchange of lithium ions with sodium ions. This study provides insights into optimizing charge storage materials for better performance in battery applications. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Advances and Perspectives of Electrolyte Modification Strategies for Vanadium-Based Materials in Aqueous Zinc-Ion Batteries** — score: 0.800 This review explores electrolyte modification strategies to overcome challenges faced by vanadium-based cathodes in aqueous zinc-ion batteries, such as dissolution and byproduct formation. It details innovative techniques like solute and solvent engineering, alongside machine learning applications, to improve electrolyte performance and enhance the stability and efficiency of these energy storage systems. Future directions emphasize the role of artificial intelligence and sustainable practices in advancing the development of these batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Unravelling anion-dominated interfacial structure of hybrid solid electrolytes via ssNMR** — score: 0.800 The article explores the interfacial structure of hybrid solid electrolytes (HSEs) using solid-state nuclear magnetic resonance (ssNMR) to identify the role of anions in enhancing ion transport across the heterogeneous phase boundaries between polymers and fillers. This understanding aims to improve the performance of solid-state lithium batteries. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Operando monitoring of mechanical failure in all-solid-state batteries via embedded optical sensing** — score: 0.800 The article discusses a novel method for real-time monitoring of mechanical failures in all-solid-state batteries through embedded optical sensing technology. This approach aims to enhance the understanding of battery performance and reliability, potentially leading to improvements in energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Interfacial engineering with cationic electrocatalysts in aqueous polysulfide redox flow batteries** — score: 0.700 This research addresses the significant power limitations in polysulfide flow batteries caused by interfacial mass transport issues. By utilizing a cationic electrocatalyst to mitigate surface repulsion, the study proposes an innovative interfacial engineering approach that boosts reaction rates dramatically, resulting in enhanced power efficiency and improved longevity for grid-scale energy storage systems. Journal: *Joule*
- **Investigation Into the Electrochemical Performance of Micron- and Nanosized Tin in Diglyme and Carbonate Electrolytes in Sodium Ion Batteries** — score: 0.700 This study evaluates the electrochemical performance of micron- and nanosized tin (Sn) electrodes in sodium ion batteries, revealing that diglyme-based electrolytes enable stable cycling despite significant volume changes, while carbonate electrolytes result in quick capacity loss, particularly with nanoparticles. The research underscores the importance of both particle size and electrolyte chemistry in enhancing the stability and performance of Sn electrodes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Dynamic Mosaic Structure SEI Derived from Boron Nitride-Graphene Oxide Nanosheets Enable Regulated Zn2+ Deposition in Aqueous Zinc-Metal Batteries** — score: 0.700 The article discusses the development of a dynamic mosaic structure solid electrolyte interphase (SEI) created from boron nitride-graphene oxide nanosheets, which facilitates controlled zinc ion (Zn2+) deposition in aqueous zinc-metal batteries. This innovative approach aims to enhance battery performance and longevity by improving charge distribution and reducing dendrite formation. Journal: *ScienceDirect Publication: Nano Energy*
- **A Comprehensive Review on Layered Oxide Cathodes for Sodium-Ion Batteries: Design,**

Manufacturing, and Commercialization — score: 0.600 This review analyzes the development of layered oxide cathodes for sodium-ion batteries, focusing on their potential for high performance and cost efficiency. It outlines the challenges in moving these materials from research to commercialization, discussing issues in design, manufacturing scalability, and market integration, while providing a strategic roadmap for industry professionals to enhance the practical application of these technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Carbon pore reorganization in seconds: Transforming activated carbon into high-plateau-capacity sodium-ion battery anodes** — score: 0.600 The article discusses a rapid process for reorganizing the carbon pore structure of activated carbon to enhance its performance as anodes in sodium-ion batteries. This transformation significantly improves the charge capacity of the battery, suggesting potential advancements in energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Revealing the effect mechanism of texture engineering on potassium-ion storage of micron-sized carbon anodes** — score: 0.600 This study investigates how texture engineering influences potassium-ion storage in micron-sized carbon anodes. The authors explore various textural modifications and their mechanisms, revealing significant enhancements in electrochemical performance that could improve the efficiency of potassium-ion batteries. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **An analytical model to describe self-discharge rates in solid-state batteries** — score: 0.500 The study develops an analytical model addressing the self-discharge rates in solid-state batteries, identifying that both electronic conductivity and electrochemical stability of the solid separator significantly influence charge loss over time. This model aims to provide insights for improving the design of battery separators and cells, enhancing their shelf life. Journal: *Nature Energy*
- **[ASAP] Size-Driven Kinetic Transitions and Strain Evolution in Bi Anode for K-Ion Batteries** — score: 0.500 The article discusses the impact of size on kinetic transitions and strain changes in bismuth (Bi) anodes used for potassium-ion batteries. It emphasizes how these factors influence the performance and stability of the anodes, providing insights that may guide future developments in energy storage technology. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **A Hydroxylated Zwitterion Enables Dual-Modal Synergy for Stable Zinc-Iodine Batteries** — score: 0.400 The article addresses the challenges of soluble polyiodides in aqueous zinc-iodine batteries, which lead to material loss and corrosion. It introduces a novel hydroxylated zwitterion that enhances stability and performance, facilitating dual-modal synergy in the battery's operation. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Dynamic localized domains of metallic glasses enable high-capacity SbBi anode for potassium ion batteries** — score: 0.400 This study explores the use of metallic glass alloys as anodes in potassium-ion batteries, highlighting their ability to mitigate issues such as volume expansion and limited capacity. The findings suggest that dynamic localized domains in these metallic glasses can significantly enhance the performance of SbBi anodes, paving the way for higher-capacity PIBs. Journal: *RSC - Energy Environ. Sci. latest articles*
- **[ASAP] Synergistic N, S-Doping Modulates the Electronic Structure of Carbon Hosts for Highly Reversible and Rapid Iodine Conversion in Solid-State Batteries** — score: 0.400 The article discusses how synergistic doping of nitrogen and sulfur alters the electronic properties of carbon materials, enhancing their performance as hosts in solid-state batteries. This modification facilitates highly reversible and rapid iodine conversion, which is crucial for improving battery efficiency. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **A Bifunctional Janus-type Solid Polymer Electrolyte Simultaneously Suppressing Polyiodide Shuttle and Accelerating Zn2+ Transport for High-Performance Zn-I2 Batteries** — score: 0.400 The article presents a Janus-type solid polymer electrolyte designed to enhance the performance of zinc-iodine batteries by simultaneously preventing polyiodide shuttle effects and facilitating the transport of Zn^2 ions. This dual functionality aims to improve battery efficiency and longevity,

addressing key challenges in current battery technologies. Journal: *ScienceDirect Publication: Nano Energy*

- **Bridging Redox Asymmetry in Hot and Cold Cells for Boosted Power Density in Thermally Regenerative Flow Batteries** — score: 0.400 The article discusses the optimization of thermally regenerative flow batteries (TRFBs) by addressing the redox asymmetry encountered in hot and cold cells, which limits their power output. The authors propose solutions to enhance the kinetic performance in these systems, potentially improving efficiency in harnessing low-grade waste heat. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Modulating the hydrogen-bond network via molecular crowding for high-voltage electrochemical stability in aqueous zinc batteries** — score: 0.400 The article discusses a novel approach to enhance the electrochemical stability of aqueous zinc batteries by modulating the hydrogen-bond network through molecular crowding techniques. This innovation promises to improve performance at high voltage, potentially advancing the efficiency and longevity of energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Localized anion-rich solvation structure in low-concentration electrolytes enabled by non-coordinating chaotrope for deep-cycling zinc metal batteries** — score: 0.400 This study investigates the role of a non-coordinating chaotrope in enhancing the solvation structure of anions in low-concentration electrolytes, which significantly improves the longevity of aqueous zinc batteries during deep-cycling. The findings aim to address the challenges of irreversible reactions that hinder battery performance, potentially leading to more sustainable energy storage solutions. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Molecular Engineering of Conjugated Polymer Cathodes via One-Pot Preparation for High-Rate and Ultra-Stable Aqueous/Seawater Zn-Ion Batteries under Harsh Conditions** — score: 0.400 This article discusses the advancement of conjugated polymer cathodes for aqueous zinc-ion batteries through a one-pot synthesis method, enhancing their performance under harsh conditions. The study emphasizes the potential of organic materials as sustainable and high-capacity candidates for battery applications, addressing challenges such as limited redox activity that have previously restricted their practical use. Journal: *RSC - Energy Environ. Sci. latest articles*
- **[ASAP] Restoring Escaped Zincate in Rechargeable Alkaline Zinc Batteries with a Seed-in-Nanoshell Design** — score: 0.300 The article discusses a novel approach to restoring escaped zincate in rechargeable alkaline zinc batteries by employing a seed-in-nanoshell design. This technique aims to enhance battery performance and longevity by effectively managing zincate dissolution, potentially leading to more efficient energy storage solutions. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **Redox-Mediated Aluminum–Air Fuel Cells With Suppressed Hydrogen Evolution for Durable Power Generation** — score: 0.300 The study presents a redox-mediated aluminum–air fuel cell (RM-AAFC) that utilizes a soluble redox mediator, 7,8-dihydroxy-2-phenazine sulfonic acid (DHPS), to enhance aluminum oxidation while suppressing hydrogen evolution reaction (HER). By employing two aluminate extraction methods, the RM-AAFC achieves significantly improved Faradaic efficiency and delivers a peak power density of 210 mW cm−2, demonstrating a viable approach for addressing problems associated with corrosion and low efficiency in aluminum-based energy systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Improving the Electrocatalytic Activity of a High-Entropy Ruddlesden–Popper Perovskite Air Electrode for Solid Oxide Cells Through Composition Regulation** — score: 0.300 The study investigates high-entropy Ruddlesden–Popper perovskite materials as air electrodes for solid oxide cells, highlighting how increasing strontium content and configurational entropy enhance electrocatalytic performance and stability. The optimal composition, resulting in a polarization resistance of 0.044 Ω cm^2, provides significant power density and stability under various environmental conditions, addressing critical challenges in solid oxide cell technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **[ASAP] Electrolyte Chemistry for Zinc Halogen Flow Batteries** — score: 0.300 The article discusses advancements in electrolyte chemistry specific to zinc halogen flow batteries, focusing on optimizing performance and efficiency. It evaluates various electrolyte formulations and their impact on battery operation, paving the way for improved energy storage solutions. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **Large-scale, mechanically robust bioinspired confined MXene nanocomposites** — score: 0.200 This review highlights the development of bioinspired confined assembly strategies for MXene nanocomposites, addressing their previously limited macroscopic properties. By minimizing void defects, these approaches enhance the mechanical strength of the composites, demonstrating their potential for scalable applications in aerospace, energy, and biomedical fields. Journal: *Nature Reviews Materials*
- **[ASAP] Progress and Prospects of Solid-State Electrolytes in Metal–Sulfur Batteries** — score: 0.200 This article reviews the advancements and future potential of solid-state electrolytes in metal-sulfur batteries, highlighting their advantages over traditional liquid electrolytes, such as improved safety and energy density. The authors discuss the latest research developments, challenges faced in commercialization, and prospective strategies for enhancing the performance of these batteries. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **[ASAP] Demethylation of Fluorine-Free Ethers to Reconcile Li+ Transport Kinetics and Oxidation Stability** — score: 0.200 The article discusses a method for demethylating fluorine-free ethers to improve lithium ion transport kinetics while also enhancing oxidation stability. This approach aims to optimize battery performance through structural modifications of electrolyte materials. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Ion correlations explain kinetic selectivity in diffusion-limited solid-state synthesis reactions** — score: 0.200 The study emphasizes the importance of cross-ion correlations in understanding the dynamics of diffusion-limited solid-state synthesis reactions, shedding light on how these correlations influence the formation of intermediate phases and product yields in powder materials. Journal: *Nature Materials*
- **[ASAP] Redox-Active La2FeNiO6–rGO Nanocomposites as Electrodes with Tuned Electrochemical Properties for Asymmetric Supercapacitor Applications** — score: 0.200 The article discusses the development of La2FeNiO6–reduced graphene oxide (rGO) nanocomposites, which exhibit enhanced electrochemical properties for use as electrodes in asymmetric supercapacitors. The study highlights the tunability of these properties, optimizing performance for energy storage applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] In Situ Small-Angle X-ray Scattering Investigation of Hydrated Membrane Nanostructures in Polymer Electrolyte Fuel Cells** — score: 0.200 The study employs in situ small-angle X-ray scattering to analyze the nanostructures of hydrated membranes in polymer electrolyte fuel cells. This investigation aims to enhance the understanding of water management and proton conduction within these systems, potentially leading to improved fuel cell performance and efficiency. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Flowability Evaluation for PVDF-Mediated Solvent-Free Li-Ion Battery Electrode Powders** — score: 0.200 This study evaluates the flowability of polyvinylidene fluoride (PVDF)-mediated, solvent-free lithium-ion battery electrode powders. The findings are crucial for optimizing the manufacturing process of battery electrodes, potentially leading to improved energy storage solutions. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Coconut-Inspired Sandwich Three-Shell-Structured Silicon/Carbon Anode for Durable Lithium-Ion Batteries** — score: 0.200 This article explores the development of a novel three-shell-structured silicon/carbon anode inspired by the coconut design, aiming to enhance the durability and performance of lithium-ion batteries. The research highlights the potential of this innovative architecture to improve battery life and efficiency compared to traditional anode materials. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*

- **[ASAP] Multifunctional -Ketoenamine Covalent Organic Framework Filler for Dendrite-Free, Flexible, High-Performance All-Solid-State Lithium Batteries** — score: 0.200 The article discusses the development of a multifunctional -ketoenamine covalent organic framework that serves as a filler in all-solid-state lithium batteries. This innovative material enhances the battery's performance by promoting dendrite-free lithium deposition while maintaining flexibility and high efficiency. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Li—Nb—O Amorphous Surface Layer to Enhance Structural Stability and Electrochemical Performance of Li-Rich Layered Cathode Oxides** — score: 0.200 The article discusses the development of a lithium niobium oxide (Li—Nb—O) amorphous surface layer that enhances both the structural stability and electrochemical performance of lithium-rich layered cathode oxides. This modification aims to improve the longevity and efficiency of lithium-ion batteries, addressing common challenges faced in energy storage applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Surface Reconstruction in High-Entropy OER Electrocatalysts: Mechanisms, Energy Consumption, and Design Principles** — score: 0.200 This article reviews surface reconstruction mechanisms in high-entropy materials (HEMs) used for oxygen evolution reaction (OER) in hydrogen production, categorizing them into four main types and discussing their effects on performance and energy consumption. It highlights the importance of reconstruction strategies for enhancing catalytic activity and durability, while also addressing challenges in controllability and stability, and offering design principles for developing efficient OER electrocatalysts with industrial applicability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*